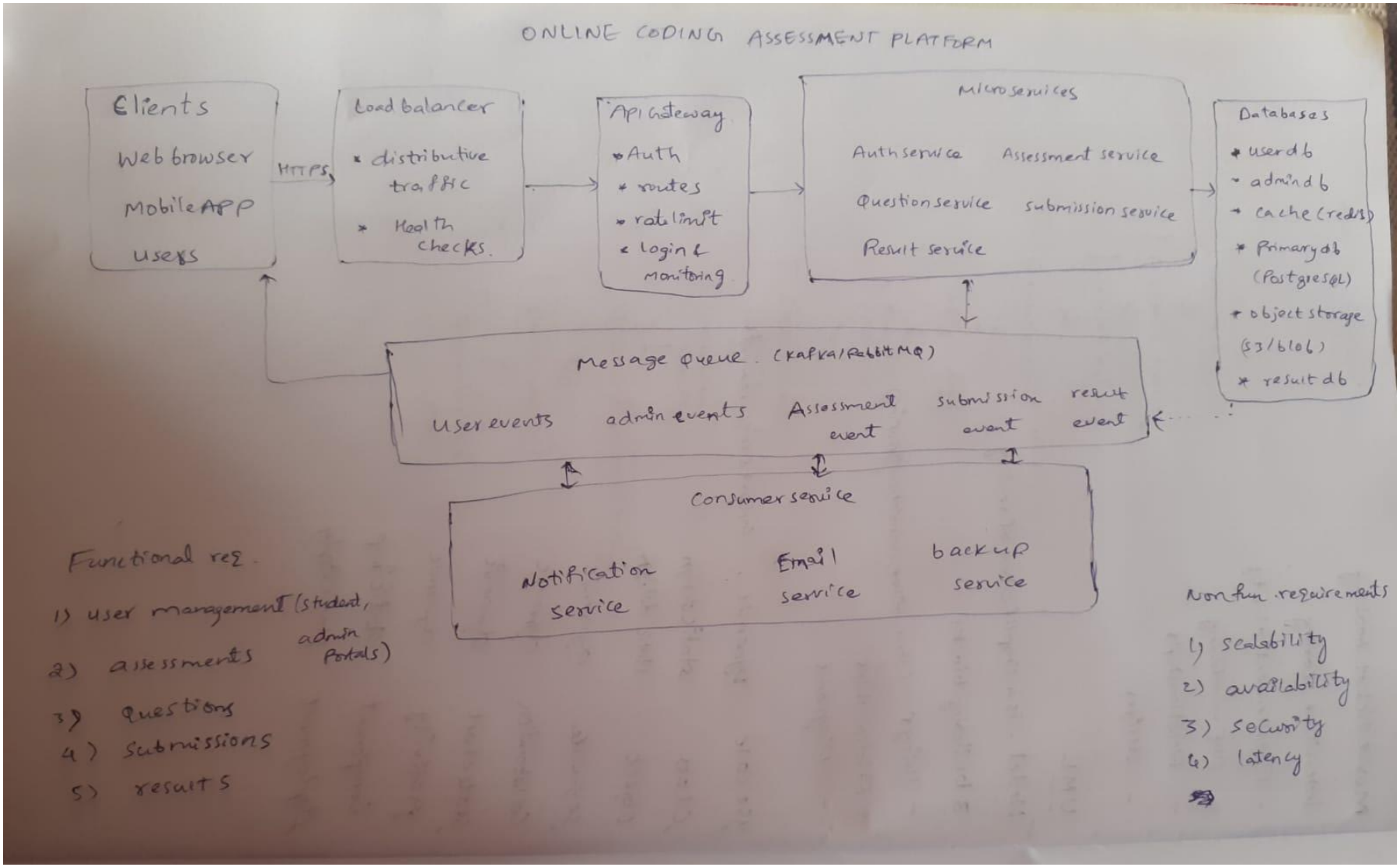

ONLINE CODING ASSESSMENT PLATFORM
Clients
Web browser
Mobile APP
users
HTTPS
Load balancer
* distributive traffic
* Health checks.
Api Gateway
* Auth
* routes
* rate limit
* login & monitoring
Microservices
Auth service
Assessment service
Question service
submission service
Result service
Databases
* user db
* admin db
* cache (redis)
* Primary db (PostgreSQL)
* object storage (S3/blob)
* result db
Message queue (Kafka/RabbitMQ)
User events
admin events
Assessment event
submission event
result event
Consumer service
Notification service
Email service
backup service
Functional req.
1) user management (student, admin Portals)
2) assessments
3) Questions
4) Submissions
5) results
Non fun. requirements
1) scalability
2) availability
3) security
4) latency